

LAB MANUAL

CSE 324

Operating Systems Lab

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Lab 3

While Loop:

```
#!/bin/bash
while [condition]
do
    command1
    command2
    command3
done
```

Script: *hello.sh*

```
#!/bin/bash
n=1
while [ $n -le 5 ]
# while (( $n <= 5 ))
do
    echo "$n"
done
```

Script: *hello.sh*

```
#!/bin/bash
n=1
while [ $n -le 5 ]
do
    echo "$n"
    sleep 3
done
```

Terminate using “[Ctrl C](#)”

Use of SLEEP:

Script: *hello.sh*

```
#!/bin/bash
n=1
while [ $n -le 5 ]
do
    echo "$n"
    n=$(( n+1 ))
# (( n++ )) post increment
# (( ++ n )) pre increment
    sleep 3
done
```

Script: *hello.sh*

```
#!/bin/bash
n=1
while [ $n -le 3 ]
do
    echo "$n"
    (( n++ ))
    gnome-terminal &
    # xterm &
done
```

For Loop:

Script: <i>hello.sh</i> FORMAT 1: <pre>#!/bin/bash for num in 1 2 3 4 5</pre> FORMAT 2: <pre>#!/bin/bash for ((EXP1; EXP2; EXP3))</pre> EXP1 - initialization EXP2 - terminating condition EXP3 - increment or decrement	Script: <i>hello.sh</i> <pre>#!/bin/bash for i in 1 2 3 4 5 do echo \$i; done</pre>
Script: <i>hello.sh</i> <pre>#!/bin/bash echo \${BASH_VERSION}</pre>	Script: <i>hello.sh</i> <pre>#!/bin/bash for i in {1..5} do echo \$i; done</pre>
Script: <i>hello.sh</i> <pre>#!/bin/bash for i in {1..10..2} # {START..END..INCREMENT} do echo \$i;</pre>	Script: <i>hello.sh</i> <pre>#!/bin/bash for ((i=0; i<5; i++)) do echo \$i; done</pre>

done	
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Until Loop:

Script: <i>hello.sh</i> <pre>#!/bin/bash until [condition] do command1 command2 commandN done</pre>	Script: <i>hello.sh</i> <pre>#!/bin/bash until [\$n -ge 10] do echo \$n n=\$((n+1)) done</pre>
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For Loop with Commands:

Script: <i>Terminal</i> <pre>for command in home pwd date do echo "New Command" \$command done</pre>	Script: <i>Terminal</i> <pre>-d -> directory for i in * do if [-d \$i] then echo \$i fi done</pre>	Script: <i>Terminal</i> <pre>-f -> file for i in * do if [-f \$i] then echo \$i fi done</pre>
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Self Study

Arithmetic Operations:

Script: <i>hello.sh</i> <pre>#!/bin/bash echo 1+1 num1=30 num2=6 echo \$((num1 + num2)) echo \$((num1 - num2)) echo \$((num1 * num2)) echo \$((num1 / num2)) echo \$((num1 % num2))</pre>	Script: <i>hello.sh</i> <pre>#!/bin/bash num1=30 num2=6 echo \$(expr \$num1 + \$num2) echo \$(expr \$num1 - \$num2) echo \$(expr \$num1 * \$num2) echo \$(expr \$num1 / \$num2) echo \$(expr \$num1 % \$num2) echo \$(expr \$num1 * \$num2)</pre>
Script: <i>Terminal</i> man bc Script: <i>hello.sh</i> <pre>#!/bin/bash num1=30.5 num2=6 echo "30.5+6" bc echo "30.5-6" bc echo "30.5*6" bc echo "30.5/6" bc echo "30.5%6" bc echo "scale=2;30.5/6" bc</pre>	Script: <i>hello.sh</i> <pre>#!/bin/bash num1=30.5 num2=6 echo "\$num1 + \$num2" bc echo "\$num1 - \$num2" bc num=27 echo "scale=2;sqrt(\$num)" bc -l # -l calls the math library where the sqrt() is defined</pre>